Paper 1

Name:

Safe speeds: fatality and injury risks of pedestrians, cyclists, motorcyclists, and car drivers impacting the front of another passenger car as a function of closing speed and age

Link:

" Safe speeds: fatality and injury risks of pedestrians, cyclists, motorcyclists, and car drivers impacting the front of another passenger car as a function of closing speed and age "

Literature Review:

This section reviews existing studies on the relationship between impact speed and injury probability for different road users. The main findings are organised by road user type.

- Pedestrians
  - Fatality risk: A 10% fatality risk occurs at impact speeds between approximately 20–60 km/h, with a best estimate of 37 km/h (Hussain et al., 2019).
  - Earlier bias: Early studies overestimated fatality risk because in-depth data samples contained more serious and fatal injuries than national data (Rosén et al., 2011).
  - Speed definition: Impact speed is typically the car's speed at impact; pedestrian speed is considered negligible.
- Cyclists

    - Fatality risk: 10% fatality risk corresponds to 60 km/h (Jeppsson & Lubbe, 2020) or 67 km/h (Rosén, 2013), based on GIDAS data.
    - Lower risk than pedestrians: Cyclists have lower injury risk than pedestrians; at any given speed, pedestrians are more likely to suffer serious injuries (Nishimoto et al., 2015).
    - Speed definition: Car impact speed is used (cyclist speed often ignored), and it gives a slightly better model fit than closing speed (Jeppsson & Lubbe, 2020).
- <u>Motorcyclists</u>
    - Fatality risk: 10% fatality risk at a closing speed of 114 km/h (Ding et al., 2019).
    - Apparent lower vulnerability: Motorcyclists seem much less vulnerable than pedestrians and cyclists, but direct comparisons are hindered by different study populations and explanatory variables.
- <u>Car Drivers</u>
    - Different metric: Most car-crash studies use delta-v (change in velocity), not impact speed or closing speed, making comparisons difficult (Bareiss & Gabler, 2020; Stigson et al., 2012).
    - Closing speed based: One study (Doecke et al., 2020) found a 10% serious injury risk at closing speeds of 71–108 km/h, depending on crash scenario.
    - Lowest vulnerability: Car occupants are among the least vulnerable road users due to protective structures (cage, airbags, etc.).

- <u>Role of Age</u>
    - Important factor: Age increases injury risk because tissues and bones stiffen with age (Schmitt et al., 2019).
    - Need to model age: Age has been successfully modelled as a covariate in injury risk curves (Bareiss & Gabler, 2020; Niebuhr

& Junge, 2017). For fair comparison across road user types, age must be explicitly included.
    - Generalisation over detail: For speed-limit setting, generalisability is more important than very specific circumstances; modelling age together with speed is feasible and beneficial.

Paper 2

Name:

Automated Construction of Time-Space Diagrams for Traffic Analysis Using Street-View Video Sequences

Link:

" Automated Construction of Time-Space Diagrams for Traffic Analysis Using Street-View Video Sequences "

Literature Review:

- Importance of Time-Space Diagrams
    - Used to analyze microscopic (headways) and macroscopic (density, flow, speed) traffic characteristics.
    - Applied in traffic control, congestion prediction, traffic state estimation, and policy making [1–9].

- Traditional Data Source: Floating Car Data (FCD)
    - Uses GPS, OBD, or cameras in vehicles.
    - Effective for estimating flow, speed, density [10–12].

- Limitation: Requires prior knowledge of penetration rate, which is hard to obtain [13].

- Camera-Based Traffic Data Collection
  - Stationary cameras (high-mount, highway) are common [14,15].
  - Recent shift to moving cameras (smartphones, dash cams, in-vehicle cameras) for Traffic State Estimation (TSE) [16–19].
  - Some studies focus on same-lane traffic [16,17]; others analyze opposite lanes [18,19].

- Gap Addressed by This Paper
  - Existing moving-camera methods focus on macroscopic parameters (flow, speed).
  - This paper aims to capture microscopic characteristics (individual vehicle trajectories) using a moving camera.
  - No need for penetration rate knowledge (unlike FCD).
  - Overcomes spatial limitations of stationary sensors.

Paper 3

Name:

Linear Regression-Based Traffic Flow Simulation: Vehicle Density and Speed Analysis on Buah Batu Road

Link:

" Linear Regression-Based Traffic Flow Simulation: Vehicle Density and Speed Analysis on Buah Batu Road "

Literature Review:

- <u>Traffic Congestion Problem</u>
  - Congestion is severe in many major cities, including Bandung, Indonesia.
  - Bandung has nearly a 1:1 vehicle-to-population ratio (approx. 2.5 million people vs. 2.2 million vehicles), causing high road density.

- <u>Traffic Flow Models – Two Main Types</u>
  - Microscopic models: Describe individual vehicle behavior (position, speed, acceleration).
  - Macroscopic models: Use partial differential equations to describe overall flow, speed, and density – known as the Lighthill-Whitham-Richards (LWR) model.

- <u>LWR Model</u>
  - Based on the conservation (continuity) equation of vehicles.
  - Useful for predicting traffic flow patterns and identifying potential congestion points based on density changes.
  - Requires a velocity-density function
  - $U = v(\rho)$
  - $U = v(\rho)$, which must be defined from observations.

- <u>Previous Studies Referenced</u>
  - Gunawan & Rizmaldi (2019) – Applied LWR model with obstacle problem on Jalan Merdeka, Bandung.
  - Harmizi et al. (2019) – Used LWR to identify congestion points.
  - Muhartini et al. (2021) – Discussed conservation equation in traffic flow.
  - These studies used macroscopic approaches, but many did not derive velocity-density functions from direct observation data using regression.

- Gap Addressed by This Paper
    - Previous studies often assumed theoretical velocity-density relationships.
    - This paper uses linear regression on real observation data from Jalan Buah Batu to derive the velocity-density function.
    - Then simulates traffic flow using the LWR model with that function.

Paper 4

Name:

Temporal instability and differences in injury severity between restrained and unrestrained drivers in speeding-related crashes

Link:

" Temporal instability and differences in injury severity between restrained and unrestrained drivers in speeding-related crashes "

Literature Review:

- Scope of Reviewed Studies (since 2010)
    - 52 previous studies on single-vehicle crash injury severity were examined.
    - They covered diverse disaggregated data: urban/rural roads, alcohol/drug impairment, number of occupants, vehicle types, weather/lighting conditions, temporal periods, etc.
    - None of these studies separately analyzed restrained vs. unrestrained drivers in speeding-related crashes while accounting for temporal instability.

- Methodological Evolution
    - Early studies used aggregate data and simple models (binary/multinomial logit).
    - Recent studies increasingly adopted random parameters (mixed) models to capture unobserved heterogeneity.
    - Advanced variants include:
    - Heterogeneity in means only
    - Heterogeneity in means and variances
    - Correlated random parameters
    - Random thresholds hierarchical ordered probit
    - These methods improve prediction accuracy, model fit, and conclusion reliability.

- Key Findings on Speeding and Seatbelt Effects
    - Speeding reduces effectiveness of occupant protection (seatbelts, airbags, crumple zones) and road safety structures (guardrails, medians).
    - Seatbelt use is endogenous (safety-conscious drivers wear belts, but some offset benefits by aggressive driving).
    - Only a few studies examined speeding at a disaggregated level (e.g., speed limit changes).
    - Seatbelt effectiveness varies by crash type, occupant position, and rear-seat belt use.
    - No prior study combined:
    - Speeding-related crashes
    - Restrained vs. unrestrained driver separation
    - Temporal instability testing
    - Out-of-sample prediction comparison

- Identified Research Gaps (Clearly Stated)
  - No study has investigated temporal instability and non-transferability of factors between restrained and unrestrained drivers in speeding-related single-vehicle crashes.
  - No study has performed predictive comparison (within-sample vs. out-of-sample) for these driver groups across time periods.

- Novel Contributions of This Paper (4 distinct points)
  - Disaggregation – Separates speeding-related crashes by seatbelt use status for the first time.
  - Temporal instability – Tests parameter stability across three biennial periods (2012–2013, 2014–2015, 2016–2017).
  - Heterogeneity modelling – Uses mixed logit with heterogeneity in means and variances to capture multi-layered unobserved factors.
  - Predictive simulation – Compares within-sample and out-of-sample predictions to quantify safety benefits of seatbelts and temporal shifts.

- Methodological Rigour Highlighted
  - Likelihood ratio tests confirmed non-transferability between restrained/unrestrained models.
  - Temporal instability tests (12 paired comparisons) showed significant parameter shifts over time – justifying period-wise analysis.
  - Out-of-sample prediction revealed: using restrained-driver parameters on unrestrained data would underestimate fatal injuries by 7–10% – demonstrating seatbelt's real safety benefit.

Paper 5

Name:

Unveiling the risks of speeding behavior by investigating driver injury severity

Link:

" Unveiling the risks of speeding behavior by investigating the dynamics of driver injury severity through advanced analytics "

Literature Review:

- Key Findings from Previous Studies:
  - Speeding is a major factor in crash severity across many countries (Thailand, US, China, Ethiopia, etc.).
  - Younger drivers (especially under 26) are significantly more likely to speed and suffer severe injuries.
  - Male drivers consistently show higher speeding behavior and severe injury risk than females.
  - Alcohol impairment dramatically increases both speeding likelihood and injury severity.
  - Roadway factors like curves, undivided roads, wet surfaces, and lack of guardrails increase severe injury risk in speeding crashes.
  - Restraint use (seatbelts) significantly reduces injury severity, but some drivers offset safety benefits by driving more aggressively.
  - Motorcycle speeding crashes are associated with specific patterns: low-volume two-lane roads, undivided curves, intersections, and wet pavement.

- Methodologies Used in Prior Work:
  - Econometric models (mixed logit, random parameters, ordered probit, logistic regression)

    - Machine learning (cluster analysis, decision trees, association rule mining)
    - Naturalistic driving data and driving simulators

- Gap This Paper Addresses:
    - Previous studies treated speeding as a binary indicator (speeding vs. not).
    - This study uses continuous speed differential (travel speed minus posted limit) as the key measure.
    - Combines machine learning (XGBoost + Random Forest with SHAP) for feature selection, then mixed logit with heterogeneity in means and variances to account for unobserved heterogeneity.
    - Uses national US crash data (CRSS 2016–2018) with out-of-sample predictions.

Paper 6

Name:

Speed and Crash Risk

Link:

" Speed and Crash Risk "

Literature Review:

- Absolute Speed & Crash Risk
    - Higher speed → higher crash rate (all studies agree).
    - The effect is exponential/power – small speed increases cause disproportionately larger crash risk.

    - Stronger on urban roads than rural roads (more interactions, complex environments).
    - Case-control studies (Kloeden et al.) are most reliable: crash risk rises exponentially with speed; no increased risk for slower vehicles.
    - Nilsson's power model (1982, 2004):
    - 2% injury crash increase per 1% speed increase
    - 3% severe injury increase
    - 4% fatal crash increase

- Speed Dispersion & Crash Risk
    - Larger speed variance (difference between fast and slow vehicles) → higher crash rate (Garber & Gadiraju, 1989; Taylor et al., 2000).
    - Older studies (Solomon, 1964) found U-shaped curves (both very slow and very fast risky). Newer studies (Kloeden) found risk only for faster vehicles, not slower.

- Main Conclusion
    - Speed is a key determinant of crash risk. Reducing speed (especially on urban roads) and reducing speed variance improve safety.

Paper 7

Name:

Crash severity analysis and risk factors (Developing country case study)

Link:

“ Crash severity analysis and risk factors (Developing country case study) ”

Literature Review:

- Common Methods Used in Previous Studies
  - Econometric models: multinomial logit, ordered logit, count regression, random parameters models.
  - Machine learning models: Decision Tree, Random Forest, Support Vector Machine, Neural Networks, XGBoost, Naive Bayes, KNN.
  - Performance: Accuracies ranged from ~60% to 99% depending on model and data.

- Key Responsible Factors Identified
  - Driver characteristics (age, gender, seatbelt use, license type, alcohol)
  - Vehicle characteristics (type, age, condition)
  - Road characteristics (surface type, classification, geometry, lane width)
  - Environmental conditions (weather, lighting, time of day, month)
  - Crash characteristics (manner of crash, number of vehicles)

- Gap Identified by the Authors
  - Most studies use official police databases, which suffer from underreporting (especially in developing countries).
  - Few studies emphasize feature selection before modeling – this can improve prediction accuracy.
  - No previous study used newspaper data as an alternative crash data source for Bangladesh.

- Contribution of This Paper

- Uses newspaper-archived crash data (441 records from 3 major Bangladeshi newspapers) as an alternate data source.
- Compares three feature selection techniques (chi-square, two-way ANOVA, regression analysis).
- Tests three ML classifiers (Decision Tree, Random Forest, Naive Bayes) with/without feature selection.
- Identifies 9 significant risk factors (day of crash, location, vehicle type, license type, seatbelt use, gender, time of crash, road surface type, road classification).

- Key Finding
  - Random Forest with chi-square or ANOVA feature selection gives the best prediction (accuracy up to 100% in some metrics).
  - Feature selection improves performance for Random Forest and Decision Tree, but not for Naive Bayes.

Paper 8

Name:

Crash severity analysis of vulnerable road users using machine learning

Link:

" Crash severity analysis of vulnerable road users using machine learning "

Literature Review:

- Traditional Statistical Approaches
  - Discrete choice models (random parameter logit, ordered probit, mixed logit) are commonly used.

    - These models effectively identify crash severity factors but have limitations with complex, non-linear relationships.

- Machine Learning Approaches (Emerging & Superior)
    - ML models handle outliers, noisy data, and missing values better.
    - No need for pre-specified mathematical functions; complex crash severity patterns can be captured.
    - Studies show ML (e.g., SVM) outperforms traditional models (multinomial logit, mixed logit) in prediction accuracy.

- ML Applied to VRU Crash Severity (Limited Previous Work)
    - Very few studies focus specifically on vulnerable road users (pedestrians, bicyclists, motorcyclists).
    - Existing VRU studies used decision trees, ensemble methods, Random Forest – but with limited explanatory variables.
    - One study on motorcyclists in Ghana found Random Forest best among classifiers; location, time, settlement type were top factors.

- Key Gaps Identified by Authors
    - No comparative analysis of VRU crash severity across individual and unified VRU groups using ML.
    - Limited number of crash parameters considered in previous ML studies.
    - No dependency analysis showing how crash severity changes with each feature (partial dependency plots missing).
    - Need to include more explanatory variables (crash type, age, gender, road/environment conditions).

- Contributions of This Paper
    - Uses 17 crash parameters (more comprehensive than prior studies).

- Compares three ML models (Random Forest, SVM, KNN) for each VRU group separately and together.
- Performs feature ranking and partial dependency analysis to show how each feature affects severity.
- Identifies similarities/differences across VRU groups.

Paper 9

Name:

Development of a Speeding-Related Crash Typology

Link:

" Development of a Speeding-Related Crash Typology "

Literature Review:

- Prior Speeding-Related Crash Typology Studies
  - Bowie & Walz (1994) – Used CARDfile, Indiana Tri-Level Study, and FARS data
  - ~70% of drivers in fatal SR crashes were in single-vehicle crashes.
  - Rural roads: 36% SR; urban: 30% SR.
  - Curves: 54% SR vs. straight sections: 27% SR.
  - Nighttime: 39% SR vs. daytime: 27% SR.
  - Male drivers: 25% SR vs. females: 16%.
  - Seatbelt use in SR fatal crashes: only 19% (vs. 37% in non-SR).
  - Alcohol involvement: 41% of drinking drivers in fatal crashes were speeding.
  - Motorcycles: >45% of fatal crashes involved speeding.

- Hendricks et al. (2001) – Analyzed 723 severe crashes (NASS CDS):
  - Excessive speed was the second leading causal factor (after inattention).
  - Primary cause for 6.8% of drivers; contributory for additional 8.1%.
  - Speeding was the leading cause of single-vehicle run-off-road crashes with traction loss and third leading cause of head-on crashes.
  - Key characteristics: curves, local/collector roads, night, clear weather, younger males (<35 years old) – males <20 made up 46.2% of the sample.

- Other Referenced Literature
  - TRB Special Report 254 (1998) – Managing speed.
  - Numerous studies on speed effects on crash frequency/severity.

- Gap Addressed by This Report
  - Previous typologies used in-depth crash investigations or smaller samples.
  - This study uses large national databases (FARS, GES) and two State databases (NC, OH) with both combined SR definition (exceeding speed limit + too fast for conditions) and over-speed-limit only definition.
  - Uses single-variable tables and CART (classification and regression tree) analyses to identify critical crash, vehicle, and driver characteristics.

Paper 10

Name:

Impact Analysis of Road Infrastructure and Traffic Control on Injury Severity

Link:

" Impact Analysis of Road Infrastructure and Traffic Control on Injury Severity "

Literature Review:

- Scope of Previous Studies (since 2014)
  - Many studies have analyzed factors affecting injury severity under limited conditions.
  - Six key factor categories identified: driver (age, sex, alcohol, seatbelt), vehicle type, environment (weather, time, day, urban/rural), accident status (rollover, airbag), road infrastructure (centerline, sidewalk boundary, alignment, lanes), traffic control (speed limit, signals, stop signs, zone 30).

- Studies on Single-Vehicle Crashes
  - Wu et al. (2016): Curved, multi-lane urban roads, alcohol, female/older drivers, rain increase severity.
  - Li et al. (2019): Curved, on-grade, multi-lane rural roads increase incapacitating injuries.
  - Zhou & Chin (2019): Median type and high-speed limits affect injury severity differently for two- vs. four-wheel vehicles.
  - Khan & Vachal (2020): Higher speed limits → more severe injuries in rollover crashes.
  - Zhang et al. (2021): High speed limits and protected left-turn signals increase severity.

- Studies on Multi-Vehicle Crashes
    - Sharafeldin et al. (2022): Urban and signalized intersections associated with lower severity; higher pavement friction reduces severity.
    - Vajari et al. (2020): Intersection-specific factors for motorcycles.
    - Yuan et al. (2022): Mixed logit models for two-vehicle crashes at unsignalized intersections.

- Comparative Studies (Single vs. Multi-Vehicle)
    - Wu et al. (2014): Significant differences in causal attributes between single- and multi-vehicle crashes.
    - Rezapour et al. (2018): Ordered logistic models for downgrade crashes.
    - Ma et al. (2022): Time, road, speed, lighting, weather correlate positively with single-vehicle severity but negatively with multi-vehicle severity.

- Studies on Road Barriers (Guardrails, Centerlines)
    - Li et al. (2018): Hitting a guardrail reduces fatal/severe injury probability; strong post-W-beam worse than low-tension cable.
    - Russo & Savolainen (2018): Severity varies across barrier types.
    - Molan et al. (2020): Rigid barriers more effective than guardrails for truck crashes.

- Gap Identified by Authors
    - Limited analysis focusing on road infrastructure and traffic control differences between single- and multi-vehicle crashes, and between intersections vs. non-intersections.
    - Most studies treat intersections and non-intersections as the same except for intercept.

- Contributions of This Paper

- Uses over 550,000 crashes in Japan (2019–2021).
- Compares single-vehicle (non-intersection only) and multi-vehicle crashes (intersection vs. non-intersection).
- Analyzes 8 road infrastructure and traffic control variables (centerline, sidewalk boundary, curve, slope, speed limit, zone 30, traffic signal, stop sign).
- Uses bias-reduced binomial logistic regression (to handle rare fatalities) and ordered logit.

Paper 11

Name:

Investigating spatial heterogeneity of speeding-related crash severity

Link:

“ Investigating spatial heterogeneity of speeding-related crash severity “

Literature Review

- Factors Related to Speeding Behavior & Injury Severity
  - Speeding behavior factors (not necessarily severity) include:
  - Driver: age, gender, driving experience, safety perception, DUI
  - Vehicle: type, power
  - Road: curve radius, grade
  - Traffic: volume
  - Weather: rain, etc.
  - Temporal: time of day
  - Injury severity in speeding-related crashes – fewer studies exist:

    - Se et al. (2023): risk factors include gender, road facility, vehicle type, crash type.
    - Yuan et al. (2023): road environment and driver age have temporally variable effects.
    - Note: Most studies combine both "exceeding posted limit" and "too fast for conditions". This study focuses only on exceeding posted limits.

- Modeling Methods for Crash Injury Severity
    - Statistical models (logistic, multinomial logit, ordered logit, nested logit)
    - Assume stable relationships across observations.
    - Can suffer from unobserved heterogeneity.
    - Advanced versions (random parameters, latent class, geographically weighted regression) allow parameters to vary.
    - Machine learning models (neural networks, random forest, etc.):
    - Superior predictive power, capturing non-linear relationships.
    - No need for assumptions like IIA.
    - Weaknesses: black-box nature, cannot address unobserved heterogeneity (including spatial heterogeneity).

- Spatial Approaches in Crash Analysis
    - Traffic crashes are geographically distributed; spatial heterogeneity (variation by location) is expected.
    - Recent studies have integrated machine learning with spatial modeling (e.g., Geographical Random Forest – Gu et al., 2023; Wu et al., 2024).
    - However, these approaches often do not adequately account for spatial dependencies among observations (unlike GWR).

- Research Gaps

- Spatial heterogeneity in speeding-related crash injury severity remains unclear.
- Existing spatial models (e.g., GWOLR) are statistical, with limitations (collinearity, explicit specification).
- Machine learning + spatial modeling attempts exist but lack proper handling of spatial weights.
- Variable importance used but does not show direction (positive/negative) of influence, limiting countermeasure targeting.

- Contributions of This Paper
  - Introduces Geographically Weighted Neural Network (GWNN) – integrates NN with geographically weighted modeling.
  - Trains a local NN model for each crash location using spatially weighted subsamples.
  - Calculates local marginal effects to show direction and magnitude of spatial variation.
  - Identifies three types of spatial variation: consistent positive, consistent negative, and inverse associations.
  - Applied to speeding-related crashes in rural Alabama (2020–2021).

Paper 12

Name:

Causal Analysis and Classification of Traffic Crash Injury Severity using ML

Link:

“ Causal Analysis and Classification of Traffic Crash Injury Severity using ML ”

Literature Review:

- Traditional Statistical Approaches
  - Binary/multinomial logit, ordered probit, nested logit, mixed models are widely used.
  - Limitations: Strict assumptions (data distribution, linear functional form), may produce erroneous estimates when violated.
  - Unobserved heterogeneity addressed via mixed models, but still constrained.

- Non-Parametric & Machine Learning Approaches (Emerging & Superior)
  - Overcome limitations of parametric models: no prior assumptions, handle non-linear relationships, noisy data, outliers.
  - Decision Trees (DT) – rule-based, easy to interpret.
  - Random Forest (RF) – ensemble of DTs, bagging, improves accuracy.
  - Support Vector Machine (SVM) – used in several studies.
  - Extreme Gradient Boosting (XGBoost) – sequential correction, fast, parallelizable.
  - Deep Neural Networks (DNN) – capture highly non-linear relationships, multiple hidden layers.

- Comparative Findings from Previous Studies
  - Harb et al. (2009) – RF used for crash avoidance maneuvers.
  - Jeong et al. (2018) – Compared logistic, DT, NN, gradient boosting, naive Bayes; bagging + oversampling improved performance.

- AlMamlook et al. (2019) – RF outperformed logistic, naive Bayes, AdaBoost.
- Chen et al. (2020) – XGBoost superior for autonomous vehicle crash severity; important features: weather, damage, location.
- Agarwal et al. (2019) – XGBoost best among CART, SVM for Indian highways; critical factors: pedestrian facility, collision type, weather, intersection, speed limit.
- Zhang et al. (2018) – RF outperformed KNN, DT, SVM, ordered probit, multinomial logit.
- Zheng et al. (2019) – CNN outperformed KNN, DT, SVM.
- Alkheder et al. (2017) – ANN better than ordered probit.
- Arhin & Gatiba (2019) – ANN high accuracy but only binary severity (injury vs. non-injury).

- Imbalanced Data Handling
  - Crash severity data highly skewed (PDO > injury > fatal).
  - SMOTE (Synthetic Minority Oversampling) used to balance classes.
  - Under-sampling of the majority class also applied.
  - Evaluation metrics: confusion matrix (true positives for rare classes) more important than accuracy.

- Causal Analysis – Limited Attention
  - Granger causality widely used in econometrics and transportation.
  - Few studies applied causal analysis to crash severity.
  - Helps identify influential variables (speed limit, weather, traffic volume, work zones, HOV lanes, etc.) and rank them.

- Gap Addressed by This Paper

- Limited research on causal inference combined with classification for crash severity.
- Most studies treat all severity levels together; this paper uses Granger causality to select features, then compares DT, RF, XGBoost, DNN for three severity classes (KA, BC, PDO).
- DNN performs best for rare KA (fatal/severe) class due to non-linear approximation.

Paper 13

Name:

Predicting and Explaining Traffic Crash Severity Through Feature Selection

Link:

" Predicting and Explaining Traffic Crash Severity Through Feature Selection? "

Literature Review:

- Limitations of Existing Crash Severity Studies
  - Most studies use small datasets (10,000–100,000 samples); only a few exceed one million (Ali et al., 2024; Behboudi et al., 2024).
  - Many suffer from limited feature diversity and lack of interpretability.
  - Data imbalance (severe crashes are rare) is a major challenge; techniques like random undersampling help (Fiorentini et al., 2020).
  - Complex feature interactions and lack of causal interpretation are key hurdles (Wen et al., 2021).

- AutoML in Crash Severity Prediction
  - AutoML automates feature engineering, model tuning, and selection.
  - Angarita-Zapata et al. (2021) demonstrated AutoML potential but noted limited integration of interpretability.
  - Baykal (2023) applied AutoML to 1.6 million crashes but lacked in-depth feature importance analysis.

- Interpretability & SHAP
  - SHAP (SHapley Additive explanations) is increasingly used for global and local interpretability (Dong et al., 2024; Cheng et al., 2024; Sattar et al., 2023).
  - Recurring predictors: collision type, contributing circumstances, vehicle characteristics; demographic factors play a lesser role.
  - Most SHAP-based studies rely on small datasets, limiting generalizability.

- Key Gaps Addressed by This Paper
  - Lack of large, feature-rich, open-access datasets.
  - Underutilization of interpretability tools within AutoML pipelines.
  - Need for scalable, transparent methodologies that handle class imbalance and complex interactions.

- Contributions of This Paper
  - Curates and publicly releases >2.3 million vehicle-level crash records (Ohio, 2017–2022) – the largest known open crash dataset.
  - Uses AutoML (JADBio) with causality-informed feature selection.

- Applies SHAP to interpret predictions and quantify feature influence.
- Identifies 17 stable predictors spanning demographic, environmental, vehicle, human, and operational categories.
- Shows that environmental and contextual factors (e.g., location, posted speed) often outweigh traditional risk factors like alcohol or distraction in model importance.